₦ MoneyPilot

Every budget is a guess until it knows local prices.

A budgeting co-pilot for Nigeria that prices a real food basket for **your** household at **this month's median in your own LGA**, then tells you what is genuinely possible — including, when it is true, that nothing is. The arithmetic is deterministic and auditable. The language model only chooses the words.

₦64,196
What the same household of four costs more, per month, in Ikeja than in Nsukka

2,207
Community price reports standing behind every figure in the product

0.00%
How far a mistyped ₦99,000 bag of rice moves the median it lands in

135
Automated tests, 739 assertions, all passing

Submission · **August 2026** Built with · **PHP, Livewire, OpenAI** Status · **Running, seeded, tested**

WHAT THIS DOCUMENT COVERS

01 The business problem
02 The working prototype
03 The solution workflow
04 Tools, platforms and AI models
05 Sample inputs and outputs
06 Evidence of testing
07 Demonstration
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01 THE BUSINESS PROBLEM

“Spend 30% on food” is a guess wearing a suit.

Every budgeting app answers with a percentage. That rule was imported from economies where prices are stable and published. In Nigeria, the same rule quietly gets one household badly wrong — and the app never finds out which one.

The same money buys different lives

A household of four on ₦260,000 needs ₦226,006 a month just to survive in Ikeja, and ₦161,810 in Nsukka. One is left ₦33,994 to live on; the other ₦98,190. A fixed percentage cannot see a 40% gap, so it prices both the same and misleads one.

There is no local price to look up

National averages are published; the price of rice in your LGA is not. So the number that decides whether a budget is honest — what food actually costs where you shop — is the one number nobody has. It has to be collected before it can be used.

A budget set in January is wrong by March

Prices move faster than plans, and nothing tells a household which line broke or by how much until the money runs out. The failure is silent, and it arrives late.

The honest answer is sometimes “it doesn't fit”

When the numbers do not work, every competitor makes them work — usually by quietly shrinking the savings goal. A budget that always balances is not telling the truth about a month that does not.

The constraint that shaped every decision: if a figure cannot be traced back to something a real person paid, it does not belong in somebody's budget. That rule produced the price network, the median that resists a typo, and the reason the language model is never allowed to decide an amount.

02 THE WORKING PROTOTYPE

Six jobs. Exactly one of them is a language model.

Running against a seeded corpus of 2,207 community price reports and 122 member households, on the same engine in every case. Every figure below was read out of the running system, not estimated.

Measures the floor	`Basket` costs 18 staples, three utilities and a real number of bus trips at the reputation-weighted median for the household's own LGA. This is the differentiator, and it uses no model at all.
Trusts the prices	`TruePrice` weights each report by recency — halving every 21 days — and by the contributor's record, then holds out anything past 2.5 median-absolute-deviations before taking the weighted median. When an LGA is too thin it widens to state, then national, and **says so on screen every time**. Currently 151 of 162 item-and-place pairs answer locally, 11 widen, none go dark.
Builds the budget	`Allocator` discounts irregular income, takes obligations off the top, funds the measured floor, then splits only the remainder by the member's own priorities. Deterministic, sub-millisecond, identical every run.
Says no, then does the work	When a month does not balance, `TradeoffService` produces concrete options priced in naira and ranked by how much each one actually hurts. Nothing is applied until the member chooses it.
Reads a bank statement	`HealthCheck` sniffs the columns from the file — GTBank, Access, Zenith, UBA, First Bank, Kuda, OPay, PalmPay — and looks first for a *timing* problem rather than an overspending one. The file is read in memory and deleted in the same request; only derived findings are stored.
Answers questions	`CoachService` matches intent and extracts entities deterministically, then answers from the same engine that built the budget. Ask twice, get the same figures. **This is the only place a model appears**, and only to rephrase an answer already decided.

The AI is an accelerator, never a dependency. Remove `OPENAI_API_KEY` and nothing about any answer changes — only how it is worded. Every figure, every budget, every trade-off and every health-check finding is computed in PHP. 16 tests hold that to account against a faked API.

03 THE SOLUTION WORKFLOW

From a household to a defended number.

1 **The member describes their situation**

Income and how reliable it is, fixed commitments, household size, where they shop. Irregular income is budgeted at a discount, because under-promising is the safe direction to be wrong in.

2 **Prices resolve for their own LGA**

Every item is priced from community reports near them. If the local data is too thin the scope widens — and the widening is labelled on screen, never passed off as a local figure.

3 **The survival floor is measured, not assumed**

18 staples, three utilities and the household's real commute, costed at this month's medians. Two people on identical incomes in different cities get materially different food allocations.

4 **The budget is built in a fixed order**

Obligations off the top, then the measured floor, then goals, then whatever is left split by the member's priorities. Shares are apportioned by largest remainder, so a budget sums to the income printed above it exactly.

5 **If it does not fit, it says so**

The shortfall is stated in naira. No goal is shrunk to make the arithmetic work. Then costed trade-offs are produced, ranked by naira recovered against how much each one costs the member in real terms.

6 **Skipping a goal is offered last, on purpose**

It frees the most money in one move, so on raw score it floats to the top and becomes the headline advice. It is ranked behind everything else by policy. Options worth under ₦500 a month are dropped rather than listed.

7 **Only then does a model touch the words**

The finished answer and the finished figures go to the model. The question does not. Any rewrite that drops, alters or invents a naira figure is discarded in favour of the engine's own wording.

THE ORDERING IS THE ENTIRE SAFETY ARGUMENT

The engine decides

Deterministic arithmetic over transactions and price reports. Ask the same question twice and get the same naira figure.

↓ hands over the answer, never the question

The model phrases

Given the finished answer and the finished figures. It cannot decide anything, because it is never told what was asked.

↓ every ₦ figure counted, in and out

The rewrite is checked, then kept or thrown away

A dropped figure is a rejection. An invented figure is the more dangerous direction — money the engine never authorised, in the engine's voice — and is also a rejection.

↳ verified live: gpt-4o-mini, 7 of 9 rewrites accepted, 2 rejected and replaced by engine wording

04 TOOLS, PLATFORMS AND AI MODELS

What it is made of.

AI MODELS — ONE JOB, DELIBERATELY

JOB	MODEL	WHY THIS ONE
Phrasing an answer	`gpt-4o-mini`	It is handed a finished answer and asked to say it more naturally. Rewriting two sentences needs no frontier model, and this runs on every question a member asks.
Every figure	`no model`	Medians, baskets, allocations, trade-offs, statement findings and the four dashboard detectors are all arithmetic. Free, instant, reproducible, and auditable months later.
Understanding a question	`no model`	Intent patterns and entity extraction in code. Financial advice that cannot be reproduced cannot be audited.

↳ `config/services.php` · swappable endpoint, no vendor SDK in any screen or job

PROGRAMMING LANGUAGES

LANGUAGE	USED FOR	SIZE
PHP 8.3	Engine, services, screens, commands, tests	13,652 lines
— of which, engine	`app/Domain` — no Eloquent, no request, no container	2,318 lines
— of which, tests	Unit and feature, against the real seeded database	2,244 lines
Blade	Member app, operations console, marketing site	5,528 lines
CSS	One hand-written design system, no build step	1,970 lines
JavaScript	Icon hydration and theme only — plus the screenshot harness	764 lines
SQL	23 tables via migrations	—

PLATFORMS AND SERVICES

Laravel 13.26	Framework, queues, scheduler, policies
Livewire 4.4	Server-rendered screens with no front-end build
MySQL 8	Single system of record. Every money column is `bigint` kobo — no floats anywhere
OpenAI API	Phrasing only, behind a switch that defaults to off
Resend	Transactional email; welcome mail is queued so SMTP never blocks sign-up
Laravel Cloud	Hosting, managed queues, hourly `prices:refresh`
PHPUnit	135 tests, 739 assertions
IBM Plex Sans / Mono	Self-hosted, including the Latin-Extended range the ₦ sign lives in

Two rules run through the whole schema. Every money column is integer kobo, because `0.1 + 0.2 ≠ 0.3` and in a budgeting app that drift shows up as allocations that do not sum to income. And transactions and price reports are append-only: a correction is a new row pointing back at the original, which costs one row and buys the ability to answer “why did my budget change?” months later.

05 SAMPLE INPUTS AND OUTPUTS

Real exchanges, taken from the running system.

A. The median that survives a typed zero

INPUT · SIX HONEST REPORTS FOR RICE, ONE FAT FINGER

₦2,250 · ₦2,300 · ₦2,280 · ₦2,320 · ₦2,260 · ₦2,290 · **₦99,000**

OUTPUT

The median stays at **₦2,280**. The outlier is held out of the aggregate and flagged for a human — never deleted, because a price that looks wrong is sometimes a market that moved.

`median moved 0.00%` `7 reports, 6 used` `a mean would have moved +605.1%`

↳ App\Domain\TruePrice::aggregate() · 2.5 MAD hold-out, 21-day half-life

B. Asking whether something is affordable, when it is not

INPUT · AMAKA, IKEJA, HOUSEHOLD OF FOUR

Can I afford ₦8,000 for dinner tonight?

THE ENGINE DECIDES

Your month already does not balance — you are short ₦99,672 before this. Spending ₦8,000 now means borrowing it from something already committed.

`Left for food: ₦44,103` `Left day-to-day: ₦72,007` `Days remaining: 4` `After this spend: ₦64,007`

`Goals protected: ₦46,667`

THE MODEL PHRASES · EVERY FIGURE CHECKED

Your budget is already off balance — you're short ₦99,672. If you spend ₦8,000 now, you'll be borrowing from something you've already committed to.

C. A question the product refuses to answer

INPUT

Should I buy bitcoin?

OUTPUT · NEVER SENT TO THE MODEL AT ALL

I do not give investment advice — that is a regulated thing and I am a budgeting tool. I can tell you what you can afford this month, what things cost near you, and where your money is going.

`intent: refused` `zero API calls`

↳ test_a_refused_question_is_never_sent_to_the_model_at_all

D. Being told the month does not fit — and what to do about it

INPUT · AMAKA IS SHORT ₦99,672 THIS MONTH

What should I do?

1	Give “Emergency fund” 12 more months	+₦31,111	closes 31%
2	Cut 4 trips out of the month	+₦4,000	closes 4%
3	Skip “Emergency fund” this month **— offered last, on purpose**	+₦46,667	closes 47%

↳ App\Services\TradeoffService::rank() · skipping the goal recovers the most and is ranked last by policy

E. The finding a pie chart cannot produce

INPUT · A CSV BANK STATEMENT, COLUMNS SNIFFED FROM THE FILE

119 transactions · 6,473 bytes · read in memory, deleted in the same request

OUTPUT · RANKED ABOVE EVERY SPENDING FINDING

Your rent leaves 24th, 4 days before you are paid. ₦85,000 goes out on about the 24th while money does not arrive until roughly the 28th. That means you carry the whole cost through the thinnest part of the month, every month. This is a timing problem, not an overspending problem — and the fix is a date, not a diet.

no narration stored · no counterparty stored · no balances stored

↳ test_no_statement_narration_reached_the_database_at_scale

F. Where something is cheapest, with its evidence

INPUT · CHIDI, LAGOS MAINLAND

Where is rice cheapest?

OUTPUT

Tejuosho Market at about ₦2,250 per kg. That is ₦50 less than the dearest market nearby.

Tejuosho ₦2,250 · 2 reports · Oyingbo ₦2,250 · 2 reports · Sabo ₦2,300 · 2 reports

06 EVIDENCE OF TESTING

What has actually been checked.

135
Automated tests

739
Assertions

21/21
Pages returning 200, browser-verified

0
Known failures

Tests run against the real seeded database — the actual price corpus, the actual moderation queue, the actual member cohort — not against mocks. The engine in `app/Domain` has no framework dependencies at all, which is what lets 19 unit tests hold it to hand-checked scenarios with no database.

SUITE	COVERS	TESTS
EngineTest	Median, MAD, decay, apportionment, basket, allocator	19
EmailTest	Verification, reset, welcome — queueing, throttling, cross-account safety	19
LanguageModelTest	The AI seam: the guard, the ordering, the key, the request count	16
ScaleTest	Query counts stay flat as the cohort grows	16
HealthCheckTest	Column sniffing, day-first dates, timing analysis, what is never stored	13
CoachTest	Intent, entity extraction, throttling, thread persistence	10
SpendTest	Append-only ledger and reversal	7
ScreensTest	Every screen renders; role gates hold	7
PriceNetworkTest	Reputation is earned, outliers are held	6
TradeoffTest	Ranking, the ₦500 floor, the goal-skip policy	6

SPEED, MEASURED WITH THE FULL COHORT LOADED

SCREEN	TIME	QUERIES
Member dashboard	161 ms	54
Prices	147 ms	15
Moderation queue	103 ms	10
Members (122 rows)	373 ms	9
Owner dashboard	82 ms	18

↳ `php artisan perf --warm · 21 pages, all HTTP 200, slowest 373 ms`

VERIFIED IN A REAL BROWSER, NOT JUST IN THE DOM

A passing DOM test tells you the markup rendered. It does not tell you that a price row wrapped to four lines on a 430px phone. A headless harness signs in through the real form, walks every screen as four different roles, and reports console errors, collapsed cards, horizontal overflow, broken images and unhydrated icons. It is the pass that caught eight `<img>` tags 404-ing on the homepage, because `asset('img/')` silently strips the trailing slash.

WHAT THE TESTS DELIBERATELY TRY TO BREAK

Access	A member cannot reach the console; only the owner sees the business dashboard. Both are server-side policy gates — signing in as an operations user and visiting `/console/business` returns 403, not a hidden link.
Credentials	Sign-in is throttled per email and IP after five failures. The same message is returned for an unknown address and a wrong password, so the form cannot be used to discover who has an account.
Privacy	No narration, counterparty, balance or individual transaction from an uploaded statement can be found anywhere in a stored row — asserted at scale, not on one example.
The model	The question never appears in the request body. A rewrite that moves a decimal point is discarded. A rewrite that *invents* a figure is discarded — the more dangerous direction, since that is money the engine never authorised, in the engine's voice.
The key	Never logged, never returned in a response, never present in rendered HTML.

What has not been proven. The price corpus is seeded — realistic community reports rather than live submissions from the public — so the medians are structurally sound but not yet market-truth. Marking that distinction on screen is done; replacing the corpus with real contributions is not. And the four dashboard detectors have been checked against seeded households, not against a person's real month. Both are named here rather than left for a judge to find.

07 DEMONSTRATION

Two minutes, four accounts.

Ten accounts are seeded, all with the password `moneypilot`, genuinely checked and rate-limited. Each one is a real row the engine built a real budget for — whether a month balances is something it worked out, not a field somebody set.

SIGN IN AS	WHERE	OPENS ON	WORTH SEEING
`chidi@student.unilag.ng`	Lagos Mainland, 1	`balances ₦5,061`	Start here. Tight but working.
`amaka.eze@gmail.com`	Ikeja, 4	`short ₦99,672`	The headline demo — costed trade-offs, goal offered last.
`sadiq.chukwu58@example.ng`	Nsukka, 5	`short ₦6,257`	One trade-off closes it. The best account for pressing **Apply**.
`t.bakare@moneypilot.ng`	Owner	`console`	62 of 122 members told, honestly, that they are short.

THE WALKTHROUGH

1. **Open the homepage and change the two cities**
   The comparison recomputes the survival floor live. Ikeja ₦226,006 against Nsukka ₦161,810 — the thesis, provable in one interaction, before signing in.

2. **Sign in as Amaka**
   The dashboard leads with ₦99,672 short and, underneath it, what is still safe to spend today — because the member whose month is hardest is the one who most needs a number to steer by.

3. **Open the trade-offs**
   Three costed options. Note that skipping the savings goal recovers the most and sits at the bottom.

4. **Sign in as Sadiq and press Apply**
   The goal's deadline moves, the budget rebuilds, and the month balances. The arithmetic genuinely changes — it is not a confirmation message.

5. **Ask the assistant anything, twice**
   Same figures both times. Every answer carries its own provenance line: *computed by the engine, phrased by a language model* — or just *computed by the engine* when the rewrite was rejected.

6 **Try to reach the console as a member**

403. Then sign in as the operations user and open the owner's dashboard: 403 again. Access is a server-side gate, not a hidden link.

The part worth watching is the refusal. Ask it about crypto and it declines, and no request leaves the building. Ask it something it can answer and every naira in the reply traces back to a transaction or a price report. That combination is the product, and it is the part a slide cannot make.

08 EXPECTED BUSINESS IMPACT

What changes for a household that uses it.

40%
The budgeting error a percentage rule makes between two Nigerian cities

18
Staples priced per household, per month, at local medians

₦0
What the arithmetic costs to run — no model touches a figure

62/122
Members told plainly that this month does not fit

For the household	A budget with a floor underneath it that is measured rather than assumed, and a single number to steer by each day. When the month does not work, they are told so in naira and handed priced options — instead of a plan that silently ate the savings goal to look tidy.
For the contributor	A price reported in thirty seconds joins the median for their own area and improves everyone else's budget. Reputation is earned rather than assumed, and an outlier is held for review rather than deleted — so contributing is worth doing and hard to poison.
For the network operator	Coverage and blind spots area by area, a moderation queue that ranks by how wrong a report is rather than by direction, and contributor standing. Currently 93% of item-and-place pairs answer locally, 11 widen and are labelled, none go dark.
For anyone auditing it	Every figure traces to a transaction or a price report, months later. Append-only ledgers mean a correction is a new row pointing at the original, so “why did my budget change?” is always answerable — which is the question that decides whether a financial tool can be trusted at all.

WHY THE ECONOMICS WORK

The expensive part of most AI products is the part MoneyPilot does not do. Medians, baskets, allocations, trade-offs and statement analysis are arithmetic, so they cost nothing and are exactly right every time. A model is called once per question, on a cheap model, purely to improve wording — and if that call fails, is slow, or is switched off entirely, the member still gets the same answer in the engine's own words. The cost of the product scales with conversations, not with households.

Honestly outstanding before a real roll-out: the price corpus needs to be real contributions rather than seeded ones, and the four dashboard detectors need measuring against households whose actual months are known. Neither is a rebuild. Both are named here because a budgeting tool that is confidently wrong about what food costs is worse than no tool, and that is the risk worth being loudest about.
